

Summer Training Program WEEK 1 - DAY 1

Topics discussed:

- General python introduction
- Introduction of instructor and company
- Nature of coding problems
- MVC architecture

Nature of coding problems:

Coding is generally problem oriented and each problem is mathematical in nature.

While solving such problems, we need to keep in mind about some things:

1. Time Complexity:

The problem should be solved in the least possible time. $[O(1)]$

2. Space Complexity:

The solution should take as little amount of memory space as possible.

3. The scale of problem:

The scale of the problem should be large for it to be profitable, looking at it via an entrepreneurial perspective, i.e. a large number of people must seek a solution to this problem. (high requirement)

MVC Architecture:

It is an industrial standard approach for software development.

It is used in the planning phase and it divides the software agent/ solution into 3 main components:

1. MODEL:

It contains all the data associated with the problem, the type of data structure and what is the best data structure for that problem. [Interview question]

2. VIEW:

It is associated with nothing but how the solution appears to the user, i.e. the user interface and user experience. It should be simple. Many companies require a dashboard which gives an instance of all the major data (may include financial report) at the same time in a very efficient fashion. View may also be associated with the general UI/UX of Applications. This is all included in the VIEW part of the MVC architecture.

3. CONTROLLER:

This is the most important part and deals with the logic of the solution. Mainly what algorithms are used to perform the required operations on data (contained in model).

It is the main part which takes up most of the computational and brain power. Fetching of data/information is also an algorithmic process.

Class example of MVC architecture:

1. UBER cab services:

a. MODEL:

It may contain following data:

Location of cab

Location of user

Type of cab

Price and other details of cab

And more

b. VIEW:

The user interface of the app shows all this information and allows users to choose and book a cab ride.

c. CONTROLLER:

It contains the computation of following:

Nearest cab to the user

Route with the least traffic

Shortest Route

Available Cabs

Acceptance/rejection by cab driver

Filters and sorting on the basis of cab price and type.

CLASS TASK:

Write the MVC architecture of a simple problem.

1. My answer

Problem: Most research engines like google scholar are just databases of a large amount of papers that search on the basis of keywords and not deep analysis, we can develop an agent that analysis the actual contents of the file, note down the keywords, limitation, main discoveries to make the world of research more efficient.

Research paper analysis agent:

a. MODEL:

Titles

Authors

Publications

Projects

b. VIEW:

Citation network

Research Gaps

c. CONTROLLER:

To perform deep paper analysis to find and keep in check of the most used keywords.

Analyse frequently occurring limitations to identify unexplored areas (research gaps).

Analyse the research impact and future possible developments.

PEER EXAMPLES:

1. Travel agent
2. Investing Mnager
3. Bed availability in hospitals
4. Buddy finder

NOTE: Agentic layer disrupts the MVC architectural approach since the agent does all the work for you. Companies like Blinkit and zomato are already working on an agentic layer where the user would do only as much as writing “i want to order this product” and the agent will place the order and process the payment just in a few clicks.

ASSIGNMENT : Identify a real time problem and sketch out the MVC architecture for the same

Problem: Surveys in vehicle insurance companies are handled manually, we can develop an agent to perform the surveys.

1. Model:

- Policy Details
- Claim Details
- Surveyor Details
- Past claims
- Pictures of the damaged vehicles

2. View:

- Claim Tracker
- Document checklist
- Dashboard

3. Controller:

- Extracts information from the documents.
- Detects fraudulent claims by analysing data consistency among documents and other data such as declarations(written, audio or video).
- Auto approves simpler claims.
- Auto assigns complex claims to surveyors.

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